HOSTEL++
Schema Documentation

1. Cities Table

This table serves as a master list for all cities where hostel services are available, primarily focusing on locations within Pakistan by default.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
city_id	INT	No	None	PK, Auto increment	Unique identifier for each city.
city_name	VARCHAR(100)	No	None	NOT NULL	The name of the city.
state	VARCHAR(100)	Yes	NULL	None	The state where the city is located.
country	VARCHAR(100)	No	Pakistan	NOT NULL	The country name.
created_at	TIMESTAMP	No	CURRENT_TIMESTAMP	NOT NULL	Record creation timestamp.

2. Areas Table

This table defines specific intra-city regions or neighborhoods, making location tracking and searches easier**.**

Column Name	Data Type	Nullable	Default Value	Constraints	Description
area_id	INT	No	None	PK, Auto increment	Unique identifier for each area.
area_name	VARCHAR(100)	No	None	NOT NULL	Name of the specific area or locality.
city_id	INT	No	None	FK (cities.city_id), NOT NULL	Links the area to a parent city.
pincode	VARCHAR(10)	Yes	NULL	None	The postal code of the area.

latitude	DECIMAL(10, 8)	Yes	NULL	CHECK (-90 to 90)	GPS latitude coordinate.
longitude	DECIMAL(11, 8)	Yes	NULL	CHECK (-180 to 180)	GPS longitude coordinate.
created_at	TIMESTAMP	No	CURRENT_TIMESTAMP	NOT NULL	Record creation timestamp.

3. Institutions Table

It stores details about universities, colleges, and workplaces that students or professionals might be affiliated with near hostels.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
institution_id	INT	No	None	PK, Auto increment	Unique identifier for the institution.
institution_name	VARCHAR(200)	No	None	NOT NULL	Name of the university or workplace.
institution_type	ENUM	No	other	NOT NULL	Type: university, college, workplace, or other.
address	TEXT	Yes	NULL	None	Physical address of the institution.
city_id	INT	No	None	FK (cities.city_id), NOT NULL	City where the institution is located.
area_id	INT	Yes	NULL	FK (areas.area_id)	Specific area of the institution.
latitude	DECIMAL(10, 8)	Yes	NULL	CHECK (-90 to 90)	GPS latitude coordinate.
longitude	DECIMAL(11, 8)	Yes	NULL	CHECK (-180 to 180)	GPS longitude coordinate.
created_at	TIMESTAMP	No	CURRENT_TIMESTAMP	NOT NULL	Record creation timestamp.

4. Hostels Table

This is the core table containing primary details, contact information, and overall status for every hostel in the system.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
hostel_id	INT	No	None	PK, Auto Increment	Unique primary key for the hostel.
hostel_name	VARCHAR(200)	No	None	NOT NULL	Display name of the hostel.
owner_name	VARCHAR(100)	Yes	NULL	None	Name of the hostel owner/manager.
contact_number	VARCHAR(20)	No	None	NOT NULL, REGEXP	Primary phone number (validated).
email	VARCHAR(100)	Yes	NULL	CHECK (format)	Contact email address (validated).
city_id	INT	No	None	FK (cities.city_id), NOT NULL	City location of the hostel.
total_rooms	INT	No	0	NOT NULL, CHECK (>=0)	Total count of rooms available.
gender_preference	ENUM	No	co-ed	NOT NULL	male, female, or co-ed.
rating	DECIMAL(3, 2)	No	0.00	CHECK (0 to 5)	Average user rating.
is_verified	BOOLEAN	No	FALSE	NOT NULL	Status of hostel verification.

5. Room Types Table

A lookup table that defines the various categories of rooms available.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
room_type_id	INT	No	None	PK, Auto increment	Unique identifier for the room type.
type_name	VARCHAR(50)	No	None	NOT NULL, UNIQUE	Name of the type
description	TEXT	Yes	NULL	None	Detailed description of the room type.

6. Hostel Rooms Table

This table acts as the inventory management system, tracking individual rooms, their capacities, current bed availability, and specific rental pricing.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
hostel_room_id	INT	No	None	PK, Auto Increment	Unique identifier for each room.
hostel_id	INT	No	None	FK (hostels), NOT NULL	Links room to a specific hostel.
room_type_id	INT	No	None	FK (room_types), NOT NULL	Categorizes the room
room_number	VARCHAR(20)	Yes	NULL	UNIQUE (with hostel_id)	The physical room identifier.
total_beds_in_room	INT	No	None	NOT NULL, CHECK (>0)	Maximum capacity of the room.
available_beds	INT	No	None	NOT NULL, CHECK (>=0)	Number of currently vacant beds.
monthly_rent	DECIMAL(10, 2)	No	None	NOT NULL, CHECK (>0)	Monthly rental cost per bed.
security_deposit	DECIMAL(10, 2)	Yes	0.00	CHECK (>=0)	Refundable deposit amount.

is_available	BOOLEAN	No	TRUE	NOT NULL	Overall availability toggle.
floor_number	INT	Yes	1	CHECK (>=0)	The floor where the room is located.
created_at	TIMESTAMP	No	CURRENT_TIMESTAMP	NOT NULL	Record creation timestamp.

7. Amenities Table

A list defining all possible facilities that a hostel could offer, categorized by type.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
amenity_id	INT	No	None	PK, Auto Increment	Unique identifier for the amenity.
amenity_name	VARCHAR(100)	No	None	NOT NULL, UNIQUE	Name (e.g., WiFi, CCTV).
category	ENUM	No	basic	NOT NULL	basic, food, furniture, etc.
icon_name	VARCHAR(50)	Yes	NULL	None	UI icon identifier.
created_at	TIMESTAMP	No	Current Timestamp	NOT NULL	Record creation timestamp.

8. Hostel Amenities Table

A junction table that maps specific amenities to hostels, allowing for additional charges or specific notes per hostel.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
hostel_id	INT	No	None	PK, FK (hostels)	Part of composite PK; links hostel.
amenity_id	INT	No	None	PK, FK (amenities)	Part of composite PK; links amenity.
is_available	BOOLEAN	No	TRUE	NOT NULL	Status of amenity at this hostel.
additional_charges	DECIMAL(10,2)	No	0.00	CHECK (>=0)	Extra cost if not included in rent.
notes	TEXT	Yes	NULL	None	Specific details

9. Meal Types Table

A lookup table defining the different meal sessions available (e.g., Breakfast, Dinner) and their standard timings.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
meal_type_id	INT	No	None	PK, Auto Increment	Unique identifier for the meal type.
meal_name	VARCHAR(50)	No	None	NOT NULL, UNIQUE	Name of meal
meal_time	TIME	Yes	NULL	None	Standard serving time.

10. Hostel Food Options Table

Details the specific food services provided by a hostel, including dietary categories and pricing.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
food_id	INT	No	None	PK, Auto Increment	Unique identifier for food record.
hostel_id	INT	No	None	FK (hostels), NOT NULL	Links food plan to a hostel.
meal_type_id	INT	No	None	FK (meal_types), NOT NULL	Type of meal being offered.
food_category	ENUM	No	both	NOT NULL	veg, non-veg, or both.
is_included_in_rent	BOOLEAN	No	FALSE	NOT NULL	Whether meal is free with room.
monthly_cost	DECIMAL(10,2)	Yes	NULL	CHECK (>=0)	Cost if purchased separately.
description	TEXT	Yes	NULL	None	Menu details or special notes.

11. Nearby Categories Table

A ookup table used to categorize points of interest near a hostel, such as Hospitals, Metro Stations, or Restaurants.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
category_id	INT	No	None	PK, Auto Increment	Unique identifier for the category.
category_name	VARCHAR (100)	No	None	NOT NULL, UNIQUE	Name of the category.
icon	VARCHAR (50)	Yes	NULL	None	UI icon class or name.

12. Nearby Places Table

This table tracks specific landmarks or facilities near a hostel, including distance and accessibility metrics.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
place_id	INT	No	None	PK, Auto Increment	Unique identifier for the place.
hostel_id	INT	No	None	FK (hostels), NOT NULL	Link to the associated hostel.
place_name	VARCHAR (200)	No	None	NOT NULL	Name of the landmark/facility.
category_id	INT	No	None	FK (nearby_categories)	Type of place.
area_id	INT	Yes	NULL	FK (areas)	Area where the place is located.
distance_km	DECIMAL(5,2)	No	None	NOT NULL, CHECK (>=0)	Distance from hostel in km.
estimated_time_minutes	INT	Yes	NULL	CHECK (>0)	Travel time estimate.
walking_distance	BOOLEAN	No	TRUE	NOT NULL	Flag for pedestrian access.
landmark	BOOLEAN	No	FALSE	NOT NULL	Flag if this is a major landmark.

13. Transport Routes Table

Defines public transport lines (Bus, Metro, Train) available within a specific city.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
route_id	INT	No	None	PK, Auto Increment	Unique identifier for the route.
route_name	VARCHAR (100)	No	None	NOT NULL	Name
route_number	VARCHAR (50)	Yes	NULL	None	Route code
transport_type	ENUM	No	None	NOT NULL	bus, metro, train, or auto.
city_id	INT	No	None	FK (cities), NOT NULL	City where the route operates.

14. Hostel Transport Connections Table

A junction table mapping hostels to the specific transport routes and stops they can access.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
hostel_id	INT	No	None	PK, FK (hostels)	Links to the hostel.
route_id	INT	No	None	PK, FK (transport_routes)	Links to the transport route.
stop_name	VARCHAR (200)	No	None	NOT NULL	Nearest stop/station name.
distance_to_stop_km	DECIMAL(5,2)	No	None	NOT NULL, CHECK (>=0)	Distance to the stop.
frequency_minutes	INT	Yes	NULL	CHECK (>0)	How often rides arrive.
first_ride	TIME	Yes	NULL	None	Start of service time.

last_ride	TIME	Yes	NULL	CHECK (last > first)	End of service time.

15. Users Table

Manages authentication and profile data for students, professionals, hostel owners, and administrators.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
user_id	INT	No	None	PK, Auto Increment	Unique identifier for the user.
first_name	VARCHAR (50)	No	None	NOT NULL	User's first name.
last_name	VARCHAR (50)	Yes	NULL	None	User's last name.
email	VARCHAR (100)	No	None	NOT NULL, UNIQUE	Primary login email (validated).
phone	VARCHAR (20)	No	None	NOT NULL, UNIQUE	Contact number (validated).
password_hash	VARCHAR (255)	No	None	NOT NULL	Securely hashed password.
user_type	ENUM	No	'student'	NOT NULL	student, professional, owner, admin.
institution_id	INT	Yes	NULL	FK (institutions)	User's linked institution.
verified	BOOLEAN	No	FALSE	NOT NULL	Identity verification status.
created_at	TIMESTAMP	No	Current Timestamp	NOT NULL	Account creation time.
last_login	TIMESTAMP	Yes	NULL	None	Most recent login timestamp.

16. Bookings Table

This table tracks all room reservations, handling the lifecycle of a stay from pending request to completion or cancellation, including financial tracking.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
booking_id	INT	No	None	PK, Auto Increment	Unique identifier for the booking.
booking_reference	VARCHAR(20)	No	None	UNIQUE, NOT NULL	Auto-generated reference
user_id	INT	No	None	FK (users), NOT NULL	The user who made the booking.
hostel_room_id	INT	No	None	FK (hostel_rooms)	The specific room reserved.
check_in_date	DATE	No	None	NOT NULL	Scheduled start of stay.
check_out_date	DATE	Yes	NULL	CHECK (> check_in)	Scheduled end of stay.
status	ENUM	No	pending	NOT NULL	pending, confirmed, cancelled, completed.
total_amount	DECIMAL (10,2)	No	None	NOT NULL, CHECK (>=0)	Total cost for the stay.
advance_paid	DECIMAL (10,2)	No	0.00	CHECK (<= total)	Amount paid at time of booking.
payment_status	ENUM	No	'pending'	NOT NULL	pending, partial, completed.
cancelled_at	TIMESTAMP	Yes	NULL	None	Timestamp if the booking is cancelled.

17. Reviews Table

Stores user feedback and ratings. It includes a "verified stay" constraint to ensure reviews are linked to actual bookings or admin entries.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
review_id	INT	No	None	PK, Auto increment	Unique identifier for the review.
user_id	INT	No	None	FK (users), NOT NULL	The reviewer.
hostel_id	INT	No	None	FK (hostels), NOT NULL	The hostel being reviewed.
booking_id	INT	Yes	NULL	FK (bookings)	Linked booking for verification.
rating	DECIMAL (3,2)	No	None	CHECK (1 to 5)	Overall star rating.
cleanliness_rating	INT	Yes	NULL	CHECK (1 to 5)	Specific rating for hygiene.
comment	TEXT	Yes	NULL	None	Detailed feedback text.
helpful_count	INT	No	0	CHECK (>=0)	Count of "Helpful" votes from others.

18. Hostel Photos Table

Manages the image gallery for each hostel, categorizing photos by area (e.g., room, dining) and identifying primary display images.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
photo_id	INT	No	None	PK, Auto Increment	Unique identifier for the photo.
hostel_id	INT	No	None	FK (hostels), NOT NULL	Link to the hostel.
photo_url	VARCHAR(500)	No	None	NOT NULL	Absolute path or URL to the image.

photo_type	ENUM	No	'other'	NOT NULL	exterior, room, dining, etc.
is_primary	BOOLEAN	No	FALSE	UNIQUE (per hostel)	If TRUE, this is the thumbnail image.

19. Wishlist Table

Allows users to save hostels for future consideration.

Column Name	Data Type	Nullable	Default Value	Constraints	Description
wishlist_id	INT	No	None	PK, Auto Increment	Unique identifier.
user_id	INT	No	None	FK (users), NOT NULL	The user's ID.
hostel_id	INT	No	None	FK (hostels), NOT NULL	The saved hostel's ID.
added_date	TIMESTAMP	No	Current Timestamp	NOT NULL	When it was added.

20. Audit Log Table

A system table used to track all changes (Insert, Update, Delete) across the database for security and recovery purposes.

Column Name	**Data Type**	**Nullable**	**Default Value**	**Constraints**	**Description**
log_id	INT	No	None	PK, Auto Increment	Unique log entry ID.
table_name	VARCHAR(50)	No	None	NOT NULL	Name of the table modified.
record_id	INT	No	None	NOT NULL	ID of the affected record.
action	ENUM	No	None	NOT NULL	INSERT, UPDATE, or DELETE.

old_data	JSON	Yes	NULL	None	Data state before the change.
new_data	JSON	Yes	NULL	None	Data state after the change.
changed_by	INT	Yes	NULL	FK (users)	User ID who performed the action.

USER ROLES AND PERMISSIONS

ROLE 1: GUEST (Unauthenticated User):

Description: Users browsing hostels without logging in. They can explore options to make informed decisions before registering.
Access Level: Public Read-Only (Limited)

Can Do	Cannot Do
View hostel listings, prices, and descriptions	View owner contact information
Search by city, area, or price range	Check real-time room availability
Read public reviews and ratings	Save hostels to wishlist
View nearby places and transport connections	Create bookings
Browse hostel photos	Post reviews

Data Access: Read-only access to public tables (cities, areas, hostels limited, reviews public, nearby places)

ROLE 2: REGULAR USER (Student/Professional):

Description: Registered users actively seeking accommodation. They can book rooms, save favorites, and review past stays.
Access Level: Personal Account + Public Read

Can Do	Cannot Do
Manage personal profile and preferences	View other users' data
View complete hostel details including owner contact	Book rooms without payment
Save hostels to personal wishlist	Review without completed stay
Create, view, and cancel own bookings	Modify confirmed booking dates
Post reviews for completed stays	Access hostel owner earnings
Make advance payments online	Bypass platform messaging

Data Access: Full CRUD on own data (profile, wishlist, bookings, reviews); read-only on all public data.

Business Rules: Email verification required; one review per hostel; advance payment minimum 10%; cancellation up to 7 days before check-in.

ROLE 3: HOSTEL OWNER:

Description: Users who own or manage hostels. They list properties, manage inventory, and handle bookings.
Access Level: Own Hostels Management + Public Read

Can Do	Cannot Do
Register and manage own hostels	Modify other owners' hostels

Update room inventory and pricing	Delete hostel with active bookings
Add/remove amenities and food services	Overbook rooms
Manage nearby places and transport connections	Process refunds (admin only)
View and confirm/cancel bookings for own hostels	Access guest emails directly
Respond to guest reviews	Modify master data (amenities, meal types)

Data Access: Full CRUD on own hostels data; read-only on bookings for own hostels; read-only on master tables.

Business Rules: New hostels require admin verification; must maintain accurate availability; respond to inquiries within 24 hours.

ROLE 4: ADMINISTRATOR:

Description: Platform administrators with full system access for management, moderation, and analytics.
Access Level: Full System Access

Can Do	**Cannot Do**
Manage all users (edit, suspend, verify)	Delete own admin account
Verify/unverify hostels	Modify own role or permissions
Moderate all reviews and content	Bypass audit logging
View all bookings and process refunds	
Access system analytics and reports	

Manage master data (cities, amenities, etc.)
View complete audit logs

Data Access: Full CRUD on ALL tables including audit_log.

Business Rules: Two-factor authentication required; all actions logged; sensitive operations may require secondary approval.